down such a path, we can take a pragmatic viewpoint and define time to be that which is measured by clocks. Of course, this shifts the burden of definition to the word “clock”; here we can somewhat loosely think of a clock as a device that undergoes perfectly regular cycles of motion. We will measure time by counting the number of cycles our clock goes through. A familiar clock such as a wristwatch meets this definition; it has hands that move in regular cycles of motion and we do indeed measure elapsed time by counting the number of cycles (or fractions thereof) that the hands swing through between chosen events.

Of course, the meaning of “perfectly regular cycles of motion” implicitly involves a notion of time, since “regular” refers to equal time durations elapsing for each cycle. From a practical standpoint we address this by building clocks out of simple physical components that, on fundamental grounds, we expect to undergo repetitive cyclical evolutions that do not change in any manner from one cycle to the next. Grandfather clocks with pendulums that swing back and forth and atomic clocks based on repetitive atomic processes provide simple examples.

Our goal is to understand how motion affects the passage of time, and since we have defined time operationally in terms of clocks, we can translate our question into how motion affects the “ticking” of clocks. It is crucial to emphasize at the outset that our discussion is not concerned with how the mechanical elements of a particular clock happen to respond to shaking or jostling that might result from bumpy motion. In fact, we will consider only the simplest and most serene kind of motion—motion at absolutely constant velocity—and therefore there will not be any shaking or jostling at all. Rather, we are interested in the universal question of how motion affects the passage of time and therefore how it fundamentally affects the ticking of any and all clocks regardless of their particular design or construction.

For this purpose we introduce the world’s conceptually simplest (yet most impractical) clock. It is known as a “light clock” and consists of two small mirrors mounted on a bracket facing one another, with a single photon of light bouncing back and forth between them (see Figure 2.1). If the mirrors are about six inches apart, it will take the photon about a billionth of a second to complete one round-trip